NEWTON SCHOOL

ALGEBRA 1

Radical Expressions — Simplify, Combine & Rationalize

Name: Evelyn Yoo **Date:** 2026-05-21

Standard: CCSS.MATH.CONTENT.HSN-RN.A.2 — Rewrite expressions involving radicals using the properties of exponents.

★ Kenny's #1 Insight — TWO WAYS to solve, both correct

Try this: $4\sqrt{\frac{5}{6}} - \sqrt{\frac{3}{10}}$ → final answer is $\frac{17\sqrt{30}}{30}$. You can get there two completely different ways:

WAY 1 — Rationalize each term FIRST, then combine

$$\frac{4\sqrt{5}}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} - \frac{\sqrt{3}}{\sqrt{10}} \cdot \frac{\sqrt{10}}{\sqrt{10}}$$
$$= \frac{4\sqrt{30}}{6} - \frac{\sqrt{30}}{10} = \frac{2\sqrt{30}}{3} - \frac{\sqrt{30}}{10} \text{ LCD 30:}$$
$$\frac{20\sqrt{30} - 3\sqrt{30}}{30} = \frac{17\sqrt{30}}{30}$$

WAY 2 — Build a common $\sqrt{30}$ denominator FIRST, rationalize at the end

$$\frac{4\sqrt{5}}{\sqrt{6}} \cdot \frac{\sqrt{5}}{\sqrt{5}} - \frac{\sqrt{3}}{\sqrt{10}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$
$$= \frac{20}{\sqrt{30}} - \frac{3}{\sqrt{30}} = \frac{17}{\sqrt{30}}$$
$$= \frac{17}{\sqrt{30}} \cdot \frac{\sqrt{30}}{\sqrt{30}} = \frac{17\sqrt{30}}{30}$$

Whichever feels easier for the problem in front of you is the right method. Pick a side, but know both.

Part A — Simplify each radical expression

Combine like radicals. Rationalize any denominator. Reduce all fractions.

1) $3\sqrt{3} - 2\sqrt{12} + 4\sqrt{\frac{1}{3}}$

2) $8\sqrt{10} - 3\sqrt{40} + 5\sqrt{\frac{1}{10}}$

3) $2\sqrt{\frac{7}{2}} + 4\sqrt{\frac{7}{8}} - \frac{1}{2}\sqrt{98}$

4) $3\sqrt{\frac{5}{12}} + \sqrt{\frac{12}{5}} - \frac{1}{3}\sqrt{60}$

5) $5\sqrt{3}(\sqrt{6}+2\sqrt{8})$

6) $5\sqrt{2}(4\sqrt{8}-2\sqrt{12})$

7) $2\sqrt{49x^3}-3\sqrt{16x^3}$

8) $4\sqrt{72s^4}-2s\sqrt{200s^2}$

9) $\sqrt{\dfrac{x^2}{16}+\dfrac{x^2}{25}}$

10) $\sqrt{\dfrac{x^2}{49}-\dfrac{x^2}{121}}$

Part B — Rationalize & combine

These reward the two-way thinking. Try the way that feels easier to you.

11) $\sqrt{\frac{x^2}{a^2} + \frac{x^2}{b^2}}$

12) $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}}$

13) $\frac{3}{\sqrt{5}} - \frac{2}{\sqrt{15}}$

14) $\sqrt{\frac{2}{3}} + \sqrt{\frac{3}{2}}$

15) $2\sqrt{\frac{1}{8}} + \sqrt{\frac{1}{2}}$

16) $\sqrt{8} + \sqrt{18} - \sqrt{2}$

17) $\sqrt{20} - \sqrt{45} + \sqrt{80}$

18) $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$

19) $(2\sqrt{5} + \sqrt{3})^2$

★ 20) (The Kenny problem — solve BOTH WAYS, show both)

$$4\sqrt{\frac{5}{6}} - \sqrt{\frac{3}{10}}$$

Write Way 1 on the left and Way 2 on the right. Confirm they give the same final answer.